<u>EE 608 Applied Modeling and Optimization Project Instructions</u>

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OVERVIEW: Form a group of (up to) 3 students per team, choose a topic of your interest related to any application of optimization, and submit a final report in the form of a slide deck. Datasets can be tricky (but deceptively easy looking!), so start early if you expect to do good work.

Examples of past topics:

1. the optimal distribution of covid vaccines,
2. the best pairing of wines with foods,
3. robot search and rescue planning,
4. drone video analytics,
5. natural language processing-based mental health classifier,
6. AI-driven wireless networking,
7. spatial data analysis, and
8. geospatial data analytics.

DATASETS:

You are responsible for finding a dataset to use for your project. Places where you can look for data include: Kaggle competitions, Reddit forums have discussions on these topics.

Some helpful info:

https://www.kaggle.com/datasets?tags=15002-Optimization

https://openreview.net/pdf?id=PghuCwnjF6y

https://gray.mgh.harvard.edu/research/optimization/227-cort

What to Submit:

The format for the slide deck is as follows. Follow the order and the slide count shown here. Upload your slide deck in pdf format. One upload per team only. Choose one person's name and use that to name your pdf file. It should be named in the following format: <first-name>_<last-name>-Project.pdf

1. Title slide (title of the project, full names of all group members, GitHub link to the project code) (**1 slide**)
2. Slide 2: Contributions of each team member **(1 slide)**
3. Problem Description (high-level, plain English description of the problem; why it is important?) **(1 slide)**
4. Optimization Model (objective function, constraints; explain the rationale behind your model) **(1 slide)**
5. Proposed Solution: is it convex/non-convex?; what is the feasible set?; why did you conclude that a solution may exist; closed-form solution; numerical solution using a toolbox, etc.) **(Max 5 slide)**

6. Validating Solution: how did you test your solution is valid, e.g., what dataset was used, simulation details, theoretical analysis, what problems you faced, how did you overcome them? etc. **(max 5 slides)**
7. Key Insights: What does your solution mean for the problem? What are the key observations you made from the solutions as applied to the problem? **(max 3 slides)**
8. Major Conclusions: was your approach viable? Did you run into any difficulties such as infeasibility, numerical instability, unexpected results, etc. **(2 slide)**
9. Suggestions for Improvement and Future Work: if you had more time and resources what would you have done to solve the problem? e.g., considered a more realistic model, used a larger dataset, implemented a working prototype, launched a startup and become a billionaire! **(1 slide)**

Grading:

1. Stick to the slide limit per concept. If you go over it, points will be deducted
2. Each point from 2 through 9 in the “What to Submit” section is worth 5 points.